20 The South African plant *Dichapetalum cymosum* contains fluoroethanoic acid, FCH_2CO_2H. The pK_a values of ethanoic, chloroethanoic and fluoroethanoic acids are given in the table.

	CH_3CO_2H	$ClCH_2CO_2H$	FCH_2CO_2H
pK_a	4.76	2.87	2.57

Which statements explain these data?

1 Fluorine and chlorine are both electronegative and draw electrons away from the O–H bond in the carboxyl group.

2 Fluorine and chlorine are both electronegative and destabilise the carboxylate anion.

3 Methyl groups are electron donating and stabilise the carboxylate anion.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 2 and 3 only